The Informational Chemistry of Elements: Mapping the Periodic Table via the Hamouda Informational Gravity Model (HIGM)

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(This model was developed in the framework of theoretical physics and AI)

Abstract

The standard classification of the periodic table of elements relies traditionally on the proton number (Z) and quantum mechanical valence electron configurations. While highly descriptive, this framework lacks a unified thermodynamic link to the cosmic vacuum. This paper extends the *Hamouda Informational Gravity Model (HIGM)* from cosmological scales to the atomic scale to treat chemical elements as localized, high-density **Informational Packets**. By applying the *Hamouda Inverse Law of Density and Entropy* ($S \propto 1/\rho$), we demonstrate that atomic mass density (ρ_a) dictates localized vacuum entropy (S_v), reinterpreting electronegativity, ionization energy, and chemical reactivity as a thermodynamic drive toward informational equilibrium. Furthermore, the characteristic 9.05% Informational Boost (β=1.0905) is formalized as a localized vacuum shielding constant that governs the stability of chemical bonds. This approach provides a scale-invariant, digital-physical blueprint for chemical periodicity.

Keywords: HIGM, Informational Vacuum, Shannon Entropy, Atomic Density, Chemical Periodicity, Entropic Phase Transition.

1. Introduction

Modern chemistry describes atomic interactions via electrostatic forces governed by Coulomb's law and localized wave functions. Concurrently, modern cosmology faces a crisis in its dark sector, which the *Hamouda Informational Gravity Model (HIGM)* addresses by reframing the vacuum not as a passive void, but as an active, high-density Informational Substrate (Hamouda, 2026). HIGM has successfully demonstrated that gravity is an emergent entropic phenomenon driven by infrared information flux, effectively resolving the Hubble Tension and removing the necessity for physical Dark Matter particles.

Because the fundamental equation of Shannon information entropy is inherently scale-free, a true unified physical theory must operate seamlessly from cosmic expanses down to the atomic radius (r_a). Building upon experimental electromagnetic spectroscopy and material properties (Hamouda & Amneenah, 2024), this paper introduces a novel application of HIGM: **Informational Periodicity**. We argue that the chemical properties of elements are localized manifestations of vacuum data-density processing, where a chemical reaction is reinterpreted as an entropic optimization event.

2. Mathematical Formalism & The Atomic Scale Link

2.1 The Hamouda Inverse Law of Atomic Density

The foundational axiom of HIGM states that Information Entropy (S) is inversely proportional to the mass density (ρ) of a physical system. We define this fundamental relation by accounting for the boundary constraints of the medium via:

$$\mathbf{S} = \frac{\boldsymbol{\kappa}}{\boldsymbol{\rho}}$$

Where κ is the characteristic informational scaling constant of the medium. To translate this cosmological principle to atomic dimensions, we consider an isolated atom as a spherical boundary containing an **Atomic Informational Volume ($\mathbf{V_I}$)** defined by its effective atomic radius r_a:

$$\mathbf{V_I = \frac{4}{3}\, r_a{}^3}$$

The corresponding **Local Bulk Density ($\boldsymbol{\rho_a}$)** within this volume is a function of the total atomic mass M_{atom} (where M_{atom}= A . m_u, with A being the atomic mass number and m_u being the unified atomic mass unit):

$$\rho_a = \frac{M_{atom}}{V_I}$$

Substituting this localized bulk density directly into the core HIGM relation yields the explicit equation for the **Localized Vacuum Entropy (S_a)** within the atomic domain:

$$S_a = \frac{\kappa}{\rho}$$

When evaluated across the periodic table, an atom with a highly compact atomic radius and increasing atomic mass (such as Fluorine or Oxygen) acts as a local suppressor of vacuum entropy. The ultra-high mass density inside the atomic boundary creates a localized low-entropy informational sink:

$$\lim_{r_a \to \min, M_{\text{atom}} \to \max} S_a \to 0 \quad (\text{or } S_a \to \min)$$

Conversely, large diffuse atoms (e.g., Cesium) maintain an exceptionally low localized mass density, presenting a high-entropy profile to the surrounding vacuum substrate:

$$\lim_{r_a \to \max, M_{\text{atom}} \to \min} S_a \to \infty$$

2.2 Shannon Entropy Flux in Electron Valence Shells

The boundary interaction between the atom and the vacuum substrate occurs precisely at the valence electron radius. Following the classical Shannon entropy formulation for a normalized power or spectral frequency distribution $P(\boldsymbol{\nu})$ satisfying the normalization condition $\int_0^\infty P(\boldsymbol{\nu}) d\boldsymbol{\nu} = 1$, the localized informational density H of the atomic boundary layer is governed by:

$$H = -\int_0^\infty P(\nu) \ln P(\nu) d\nu$$

In this framework, the valence electrons represent the outermost "data bytes" running on the vacuum "hardware." The energy required to perturb these bytes—traditionally known as the **First Ionization Energy (E_i)**—is the precise thermodynamic work required to overcome the localized informational pressure (P_{info}) holding the electron bit within the suppressed entropy zone of the nucleus. This pressure is defined as the spatial gradient of the local energy with respect to the informational volume:

$$P_{\text{info}} = -\left(\frac{\partial E}{\partial V_I}\right)_S$$

Integrating this pressure across the volume change required to transition an electron to an infinite radius yields the formulation for the ionization energy as an informational decoupling event:

$$E_i = \int_{V_I}^{\infty} P_{\text{info}}\, dV = \int_{r_a}^{\infty} \left(-\frac{\partial E}{\partial V_I}\right)(4\pi r^2)\, d$$

3. Reinterpreting Chemical Periodicity and Bonding

By translating classical chemical metrics into the HIGM layer of equations, the periodic table can be mapped as a continuous topological manifold of informational gradients.

3.1 Electronegativity as an Informational Gravity Gradient

Electronegativity (χ) is no longer viewed merely as an arbitrary relative scale of electron attraction, but as a direct **Informational Gravity Pull** dictated by the spatial gradient of entropy (∇S) at the atomic boundary layer. Taking the spatial derivative of our derived atomic entropy equation with respect to the atomic radius radial vector **r**:

$$\chi \propto \nabla S_a = \frac{d}{dr}(S_a)\,\hat{\mathbf{r}} = \frac{d}{dr}\left(\frac{4\pi\kappa r^3}{3M_{\text{atom}}}\right)\hat{\mathbf{r}}$$

Evaluating the derivative directly yields the explicit vector equation for Electronegativity:

$$\boldsymbol{\chi} = \xi \cdot \left(\frac{4\pi\kappa \cdot 3r^2}{3M_{\text{atom}}} \right) \hat{\mathbf{r}} = \xi \cdot \frac{4\pi\kappa r^2}{M_{\text{atom}}} \hat{\mathbf{r}}$$

Where ξ is an empirical scaling constant matching the metric to standard Pauling units. Elements on the top right of the periodic table possess an extreme spatial gradient ∇S_a, behaving as high-pressure informational vacuums craving completion to achieve geometric structural balance.

3.2 The 9.05% Informational Boost (β) in Chemical Bonding

In cosmic structures, the 9.05% informational constant represents the entropic pressure boost that accounts for flat galactic rotation curves (Hamouda, 2026). It is expressed mathematically as the multiplier:

$$\beta = 1 + 0.0905 = 1.0905$$

At the molecular scale, this constant acts as a **Vacuum Substrate Shielding Factor**. When two atoms engage in a chemical reaction, the transfer or sharing of valence electrons constitutes an **Entropic Phase Transition**. The total energy release (ΔE_{bond}) during the configuration of a stable covalent or ionic lattice is scaled directly by the vacuum's informational impedance:

$$\Delta E_{bond} = \beta \, . \, \Delta H_{system}$$

Expanding ΔH_{system} into its constituent components yields the full predictive energy formula:

$$\Delta E_{\text{bond}} = \beta \cdot \left[H_{\text{final}} - \sum_{k=1}^{n} H_{\text{atom},\, k} \right]$$

For a binary molecular system comprising Atom 1 and Atom 2, the exact energy calculation expands to:

$$\Delta E_{\text{bond}} = 1.0905 \cdot [H_{\text{final}} - (H_{\text{atom, 1}} + H_{\text{atom}}$$

Traditional Metric	HIGM Informational Reinterpretation
Ionic Biing (e.g., NaCl)	An entropic equalizer event where a high-entropy data byte (Na) is absorbed into an ultra-low entropy sink (Cl) to maximize collective substrate density, satisfying: $\Delta\ S_{transfer} = S_{fina}l - (S_{Na} + S_{Cl}) < 0$
Covalent Bonding	The formation of a shared "Cosmic Data Bus" between nuclei, locally compressing vacuum entropy to form stable molecular geometry by minimizing the shared overlapping volume element: $H_{\text{shared}} = \int (P_1 \cap P_2) \ln(P_1 \cap P_2) d\nu$
Lanthanide Contraction	An accumulation of informational density where increasing nuclear mass outpaces volume growth, causing a catastrophic localized drop in vacuum entropy (S_a), formalized as: $\frac{\partial^2 S_a}{\partial Z^2} < 0 \quad \text{for } Z \in [57, 71]$

4. Interdisciplinary Scaling: From Nano-Systems to Bio-Chemistry

The scale-invariance of HIGM permits immediate application to advanced material systems and biology. At the nanometer scale (1nm} - 100 nm), quantum confinement changes the surface-to-volume ratio exponentially. HIGM maps this as a localized inflation of Shannon entropy. Let the surface area be $A_s = 4\pi r^2$ and the volume be $V = \frac{4}{3}\pi r^3$ The ratio is given by:

As r $\rightarrow$1nm}, this ratio increases dramatically, causing a localized inflation of Shannon entropy. The model predicts a precise boundary thermal signature as infrared information flux exports data to maintain nanoscale equilibrium, governed by the heat dissipation rate:

$$\frac{dQ}{dt} = T\frac{dS_a}{dt} = T \cdot \kappa \frac{d}{dt}\left(\frac{1}{\rho_a}\right)$$

Furthermore, this framework illuminates biological signaling. Recent evaluations of elemental concentration in the human body (Hamouda, 2026) emphasize the strict compartmentalization of Sodium (Na) and Potassium (K) across cellular membranes. Under HIGM, the brain's ion channels operate as quantum information gates. The active potential gradient across the membrane boundary is a function of the localized entropic density difference:

$$\Delta\Phi_{\mathrm{membrane}} = \gamma \cdot (H_{\mathrm{K}^+} - H_{\mathrm{Na}^+})$$

The rapid flux of these distinct informational packets shifts the local entropic pressure of the cellular substrate, serving as the underlying thermodynamic mechanism that drives neural action potentials and cognitive processing according to the time-dependent flux equation:

$$\frac{\partial H_{\mathrm{membrane}}}{\partial t} = D\nabla^2 H + \beta \cdot J_{\mathrm{info}}$$

Where D is the informational diffusion coefficient and J_{info} is the net incoming data flux.

5. Conclusion

The extension of the Hamouda Informational Gravity Model (HIGM) to the periodic table of elements successfully bridges the gap between quantum chemistry and emergent thermodynamic cosmology. By replacing purely particle-centric paradigms with an active, data-dense Informational Substrate, chemical interactions are beautifully simplified into a universal drive toward entropy optimization. The periodic table is thus revealed to be a masterpiece of cosmic software tuning, running seamlessly on the hardware of the informational vacuum.

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